



[45]

SARDAR PATEL UNIVERSITY

B.C.A. – V SEMESTER EXAMINATION 2023 (OLD COURSE)

US05DBCA21: COMPUTER GRAPHICS

Date: 30/10/2023 (Monday)

Time: 10.00 A.M. TO 01.00 P.M.

TOTAL MARKS: 70

NOTE: FIGURE TO RIGHT INDICATES FULL MARKS OF THE QUESTIONS.

Q-1 Select the appropriate option for the following.

10

- 1 CAD stands for _____.
 A. Computer Aided Design B. Common Application Design
 C. Common Aided Design D. Computer Application Design
- 2 _____ is a primary input device for entering string.
 A. Mouse B. Joystick
 C. Trackball D. Keyboard
- 3 _____ is the equation for straight line.
 A. $Y = m * x + b$ B. $Y = m / x + b$
 C. $Y = m * x * b$ D. $Y = m * x - b$
- 4 The overall design style for a set of characters is called _____.
 A. Character Style B. Face Style
 C. Typeface D. Character face
- 5 A _____ is applied to an object by repositioning it along a straight – line path from one coordinate location to another.
 A. Scaling B. Translation
 C. Shear D. Reflection
- 6 Line is considered to be completely inside if the region code is _____.
 A. 1010 B. 1100
 C. 0000 D. 0101
- 7 _____ mode application program and input device operate independently.
 A. Request mode B. Event mode
 C. Sample mode D. None of Above
- 8 Which device specifying a coordinate position(x,y)?
 A. Locator B. String
 C. Stroke D. Pick
- 9 _____ is produced by capping the connection between the two segments with a circular boundary.
 A. Round Join B. Bevel Join
 C. Square Join D. Miter Join
- 10 Beam Penetration Method displays maximum _____ colors.
 A. 8 B. 4
 C. 16 D. 32

[11]

[P.T.O.]

- Q-2 Give Short answers of the following questions (ANY TEN). 20**
- 1 Define the term: Aspect ratio, Pixel
 - 2 State disadvantage of Beam Penetration method.
 - 3 Explain Voice Systems in brief.
 - 4 List various character attributes.
 - 5 Compare and contrast: serif and sans serif.
 - 6 List attributes for the line. Explain any one.
 - 7 Define the term: Scaling, Reflection
 - 8 What is clipping? List out all clipping object.
 - 9 Explain concept of line clipping in brief.
 - 10 List various input modes.
 - 11 Explain consistency in brief.
 - 12 What is a user model?
- Q-3 List all types of input devices and explain any four in detail. 10**
- OR**
- Q-3 What is Computer Graphics? Explain in brief a major application area of Computer Graphics. 10**
- Q-4 (A) Write steps of Bresenham Line Drawing Algorithm. 05**
(B) Write steps to generate circle using Mid-Point Algorithm. 05
- OR**
- Q-4 Write a note on character attributes. 10**
- Q-5 (A) Explain in brief Cohen Sutherland Line Clipping Algorithm 05**
(B) Explain window-to-viewport coordinate window transformation. 05
- OR**
- Q-5 What is geometric transformation? List all 2-D geometric transformation and explain any two in detail. 10**
- Q-6 Explain in detail general considerations in structuring a user dialogue. 10**
- OR**
- Q-6 (A) List and explain interactive picture-construction techniques. 05**
(B) Explain purpose of graphical input functions. Also explain different input modes in brief. 05

